MIL-C-52155C
20 August 1980
SUPERSEDING
MIL-C-52155B
7 May 1975

MILITARY SPECIFICATION

CHARGES, FIRE EXTINGUISHER, ANTIFREEZE, CALCIUM-LITHIUM CHLORIDE BASE AND LITHIUM CHLORIDE BASE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers fire extinguisher antifreeze charges having either a calcium-lithium base or a lithium chloride base.

1.2 Classification. The antifreeze charges shall be the following types and sizes, as specified (see 6.2):

Type I - Lithium chloride base.

Size 1 - for 2-1/2-gallon fire extinguisher.

Type II - Calcium-lithium chloride base.

Size 1 - for 2-1/2-gallon fire extinguisher.
Size 2 - for 4-gallon fire extinguisher.

FSC 4210

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Size 3 - for 5-gallon fire extinguisher.
Size 4 - for 33-gallon fire extinguisher.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

L-P-378	- Plastic Sheet and Strip, Thin Gauge, Polyolefin.
O-C-105	- Calcium Chloride, Dihydrate and Calcium Chloride, Anhydrous: Technical.
O-S-588	- Sodium Chromate, Anhydrous, Technical.
UU-T-81	- Tags, Shipping and Stock.
PPP-C-96	- Cans, Metal, 28 Gage and Lighter.
PPP-P-704	- Pails, Metal: (Shipping, Steel, 1 Through 12 Gallon).

STANDARDS

FEDERAL

FED. TEST METHOD STD. No. 536	- Soap and Soap-Products (Including Synthetic Detergents); Sampling and Testing.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-147	- Palletized and Containerized Unit Loads, 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

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(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The charges shall consist of packaged, mixed, dry chemicals as specified herein.

3.2 First article. When specified (see 6.2), the contractor shall furnish one or more charges for first article inspection and approval (see 4.3 and 6.3).

3.3 Material. Material shall be as specified herein. Material not specified shall be selected by the contractor and shall conform to all provisions of this specification.

3.3.1 Calcium chloride. Calcium chloride shall conform to O-C-105, Type I, except the size distribution shall be as shown in Table I.

3.3.2 Lithium chloride. Lithium chloride shall be technical grade lithium chloride assaying not less than 99 percent by weight LiCl. The size distribution shall be as shown in Table I.

Table I. Size distribution for
Calcium chloride and calcium-lithium chloride

Sieve size	Amount, minimum	
	Calcium chloride	Calcium-lithium chloride
	percent by weight	percent by weight
16-mesh (pass through)	100	100
60-mesh (retained on)	30	35
100-mesh (retained on)	50	50

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3.3.3 Sodium chromate. Sodium chromate to inhibit rust shall conform to O-S-588.

3.4 Instructions and warnings. In addition to special marking specified in Section 5, each package shall carry the following information:

(a) The amount of water required for mixing (see Table III), and instructions to add the charge to water instead of the water to charge.
(b) The temperature of the water required to obtain optimum solubility time.
(c) Instructions for charging the solution into the extinguisher.
(d) Handling instructions and cautions (including cautions against electrical hazards).

3.4.1 Caution marking. All charges shall be labeled as follows:

"CAUTION: Harmful Dust: Avoid Inhalation; Avoid Prolonged or Repeated Skin Contact."

3.4.2 Warning tag. The contractor shall furnish a red tag conforming to UU-T-81, Type A, Grade 3, Size 5, with every antifreeze charge. The tag shall be printed with the following warning in prominent black letters:

"FOR CLASS A FIRES ONLY
DO NOT USE ON ELECTRICAL FIRES"

The contractor shall include instructions that the tag shall be attached to the fire extinguisher after charging.

3.5 Type I and Type II requirements.

3.5.1 Composition. When packaged, the charges expressed in pounds and kilograms to the nearest 5/100 shall be the composition specified in Table II.

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Table II. Composition of charges (as packaged)

		Calcium chloride				Lithium chloride as LiCl		Sodium chromate as Na_2CrO_4	
		as $CaCl_2$		as $CaCl_2 2H_2O$					
Type	Size	kg	lb	kg	lb	kg	lb	gm	oz
I	1	-	-	-	-	2.63	5.80	37.00	1.30
II	1	2.30	5.10	3.05	6.72	1.02	2.25	14.00	0.50
II	2	3.70	8.20	4.80	10.80	1.65	3.60	22.68	0.80
II	3	4.60	10.10	6.15	13.56	2.05	4.50	28.35	1.00
II	4	33.80	74.52	40.53	89.35	13.45	29.70	184.30	6.50

3.5.2 Performance. After the charge is dissolved at room temperature in the amount of water specified in Table III, the charge shall not freeze or produce granular crystals capable of plugging the nozzles, hose, or strainer at the test temperatures specified in Table III.

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Table III. Water volume and test temperatures

Makeup water 1/				Total volume of charge 2/		Test temperature
Type	Size	U.S. Gallons	Liters	U.S. Gallons	Liters	° F, ±2° F (° C, ±1° C)
I	1	2.16	8.18	2.50	9.46	-70 (-57)
II	1	1.98	7.49	2.50	9.46	-65 (-54)
II	2	3.17	12.00	4.00	15.14	-65 (-54)
II	3	3.97	15.03	5.00	18.92	-65 (-54)
II	4	26.17	99.05	33.00	124.90	-65 (-54)

1/ Measured at 70° F. (21° C), 1 gallon weighs 8.34 pounds (3.78 kg).
2/ Specific gravity of solution at 68° F (20° C) is 1.2265 for Type II solutions.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

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4.2 Classification of inspections. The inspection requirements specified herein shall be classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article charge shall be examined as specified in 4.5.1. Presence of one or more defect shall be cause for rejection.

4.3.2 Tests. The first article charge shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the defects specified in 4.5.1. AQL shall be 2.5 percent defective for major defects.

4.4.3 Tests. Two containers, including their contents, shall be selected at random from each lot. A lot shall consist of all material mixed as a batch and offered for delivery at one time. Each container shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The charges shall be examined as specified herein for the following defects:

Major

101. Incorrect fill of container.
102. Identification or special markings missing, illegible, or incorrect.

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4.5.2 Tests.

4.5.2.1 Chemical analysis. The components of the charge shall be tested by any standard laboratory procedure. Failure of the chemical composition to comply with Table II shall constitute failure of this test.

4.5.2.2 Size distribution. The size distribution of the sample shall be determined in accordance with FED. TEST METHOD STD. No. 536, Method 2101 for dry powders. Nonconformance to Table I shall constitute failure of this test.

4.5.2.3 Performance. Approximately 25 milligrams of the charge shall be removed and set aside for seeding. The remainder of the charge shall then be dissolved at room temperature in the amount of water specified in Table III. The solution shall be conditioned for not less than 8 hours at the test temperature specified in Table III and then shock-seeded at the test temperature with the previously removed crystals, which shall be at the same temperature as the solution. The seeded solution shall be stirred vigorously for 5 minutes, shall remain at the test temperature for 3 hours, and shall then be examined. Any evidence of freezing or crystallizing shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

103.	Materials and containers not as specified. Each incorrect material or container shall be considered one defect.
104.	Outage exceeds 1 inch for Level A.
105.	Liner not provided, fabricated, and sealed as specified for Level A.

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106. Warning tag not placed within the container as specified.
107. Type I, Size 1 and Type II, Sizes 1, 2 and 3 charges not preserved and packed in accordance with the referenced document.
108. Type II, Size 4 charges not palletized as specified when applicable.
109. Marking missing, illegible, incorrect, or incomplete.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Type I, Size 1, and Type II, Sizes 1, 2 and 3. Each charge shall be preserved in a metal can conforming to PPP-C-96, Type V, Class 2, Plan B, of a size such that, after filling, the outage shall not exceed 1 inch. The can shall be provided with a polyethylene liner fabricated from material conforming to L-P-378, Type I, in the form of extruded lay flat tubing. Material thickness shall be not less than 4 mils. The bottom closure shall be by heat sealing and the top closure by heat sealing or by a double fold drawn together and secured with a twist tie. The warning tag shall be placed within the can on the top of the sealed liner and the can lid secured.

5.1.1.2 Type II, Size 4. Each charge shall be preserved as specified in 5.1.1.1, except the container shall be a metal pail conforming to PPP-P-704, Type II, class as applicable.

5.1.2 Commercial. Commercial preservation shall be in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, Level B, or Commercial as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Type I, Size 1 and Type II, Sizes 1, 2 and 3. The filled cans shall be packed as specified in the appendix of PPP-C-96 for Level A unpackaged cans, except fiberboard containers shall not be used.

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5.2.1.2 Type II, Size 4. Unless otherwise specified (see 6.2), the filled pails shall be palletized in accordance with MIL-STD-147, Load Type IV, Bonding Means K and L, Storage Aid 6. When specified (see 6.2 and 6.4), the filled pails shall be shipped loose.

5.2.2 Level B.

5.2.2.1 Type I, Size 1 and Type II, Sizes 1, 2 and 3. The filled cans shall be packed as specified in the appendix of PPP-C-96 for Level B unpackaged cans. Fiberboard shall be Class Weather-Resistant.

5.2.2.2 Type II, Size 4. Unless otherwise specified (see 6.2), the filled pails shall be palletized in accordance with MIL-STD-147, Load Type IV. When specified (see 6.2 and 6.4), the filled pails shall be shipped loose.

5.2.3 Commercial. Commercial packing shall be in accordance with MIL-STD-1188.

5.3 Marking. In addition to any special marking required by the contract or order, cans, pails, shipping containers, and pallets shall be marked in accordance with MIL-STD-129 for military levels of protection. Commercial marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The charges are intended to be used to prevent freezing in water-type fire extinguishers. The charges should not be used in extinguishers with tanks fabricated of corrosion-resisting steel.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type and size of charge required (see 1.2).
(c) When first article is required for inspection and approval and number of charges required (see 3.2, 4.3, and 6.3).
(d) Degree of preservation and degree of packing required (see 5.1 and 5.2).
(e) When palletization is not required (see 5.2.1.2, 5.2.2.2, and 6.4).

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6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of paragraph 7-104.55 of the Defense Aquisition Regulations. The first article should be a first produced charge. The first article should consist of one or more units. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, test, and approval of the first article.

6.4 Palletization. The Type II, Size 4 charges should be shipped loose without palletization only when the quantity procured is insufficient to utilize a pallet.

Custodians:
Army - ME
Navy - YD
Air Force - 99

Review activity:
DLA - CS

User activity:
Navy - MC

Preparing activity:
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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-C-52155C Charges, Fire Extinguisher, Antifreeze, Calcium-Lithium Chloride Base & Lithium Chloride Base

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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